

Introduction to Human Physiology

Physiology | Source: 01_introduction_to_Physiology.pdf (24 slides) · References: Guyton & Hall 13th ed. (Unit V, ch. 25); Ganong 25th ed. (Section I, ch. 2)

Legend: ★ = high-yield / likely exam target · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

This is a short foundation lecture (≈ 900 words of slide text). The notes stay close to it; anything beyond the slides is marked.

1. Title / Objectives

Objective (slide)	Likely tested as
Define human physiology	One-line definition; levels of organisation
★ Identify the body compartments	60–40–20 volumes, ionic composition, which marker measures which compartment
★ Describe homeostasis and its types	SRACEER components; negative vs positive feedback examples

Lecture flow: Human physiology (definition, levels of organisation) → Body compartments (composition, TBW, volumes, ions, measurement) → Homeostasis (definition, components, feedback types) → Conclusion → Case.

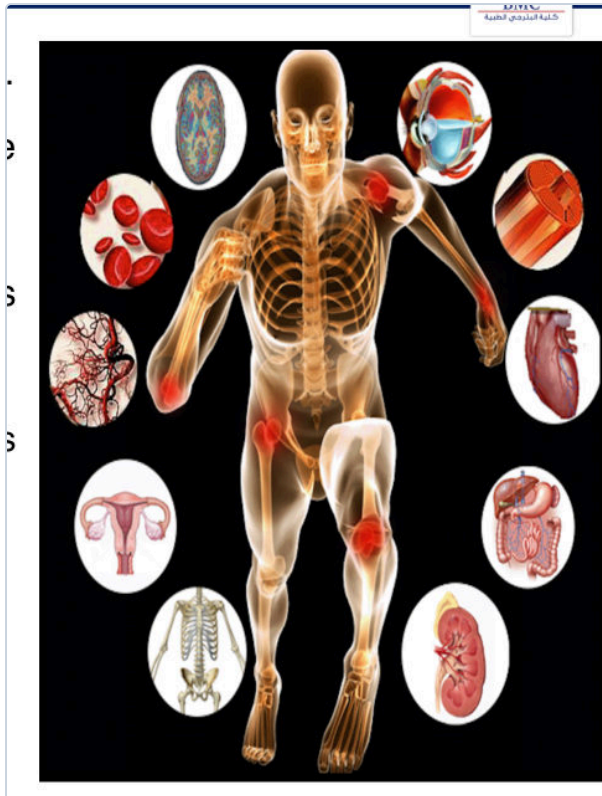
2. Main Content (Lecture Order)

2.1 Human Physiology (Slides 4–6)

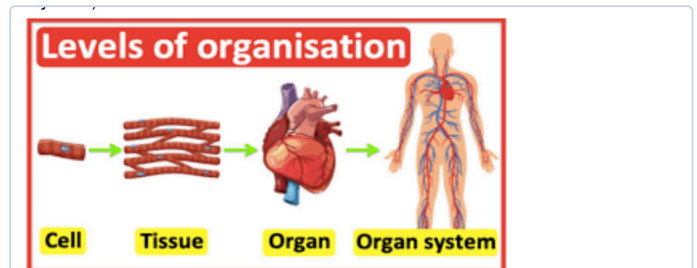
- ★ **Physiology** = the study of the **normal function of living things**
- Human physiology** = how the **human body works**: the functions of **organs and systems and their integration**
- ★ The body has **75–100 trillion (10^{12}) cells** *[note: 1 trillion = 10^{12} , so 75–100 trillion $\approx 10^{14}$]*

★ Levels of organisation (Slide 6):

- Cells**
- Tissues**: cells with similar properties (e.g., muscular tissue)
- Organs**: different tissues combined (e.g., heart, brain, liver)
- Systems**: organs with complementary functions (e.g., cardiovascular, nervous)



Slide 5: The human body as integrated organ systems



Slide 6: Levels of organisation: cell → tissue → organ → organ system

🔄 **Recap: Physiology.** Physiology is normal function. The body is 75-100 trillion cells organised into tissues → organs → systems that work together.

🔗 **Link:** Every one of those trillions of cells lives in fluid. To understand function, first know where the body's water is and what it contains.

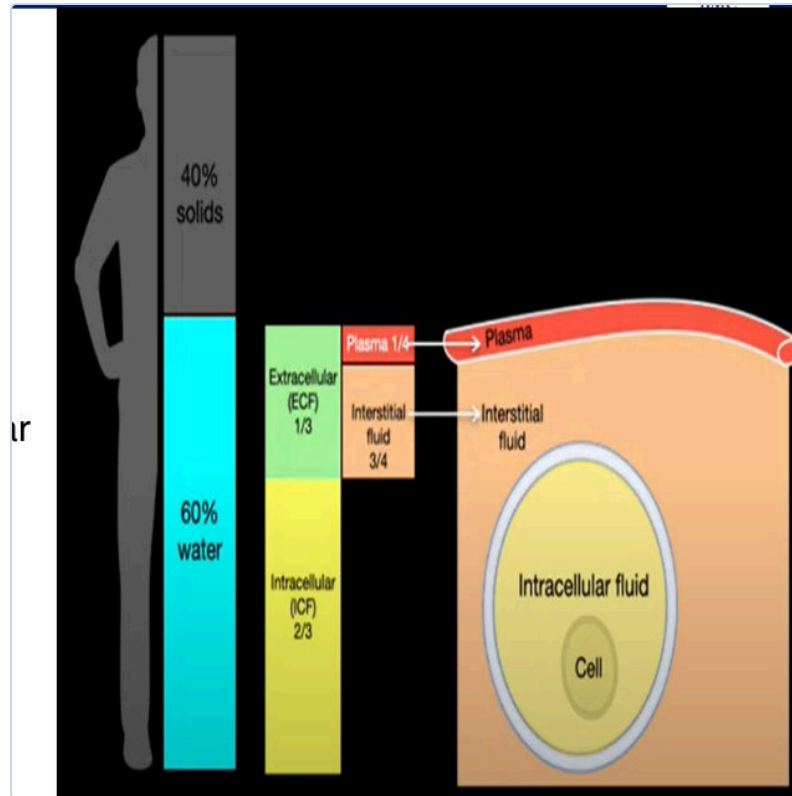
▶ **Up next: Body compartments.**

2.2 Body Compartments (Slides 7-13)

A. Body composition (Slide 8), average adult male

Component	% body weight
★ Water	★ 60%
Protein & related substances	18%
Fats	15%
Minerals	7%

The body is divided into **intracellular (ICF)** and **extracellular (ECF)** compartments.

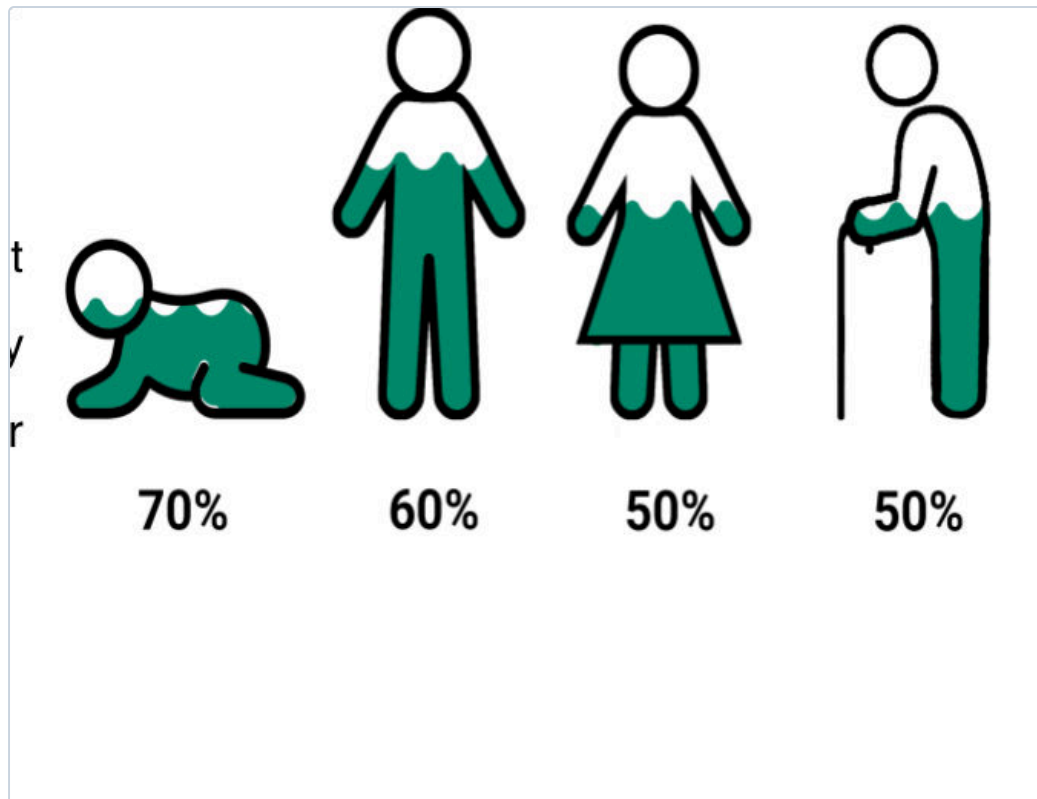


Slide 8: Body composition: 60% water, 40% solids

B. ★★ Total body water (TBW) varies with age, sex, fat (Slide 9)

Group	TBW (% body weight)
★ Infants	★ 70-75%
★ Adult male	★ 60%
★ Adult female	★ ~54% (slide figure shows ~50%)
Obese	< 60% (varies with obesity type)
Elderly	~50% (slide figure only)

★ **Why lower in women and the obese: high fat content** (fat holds little water).

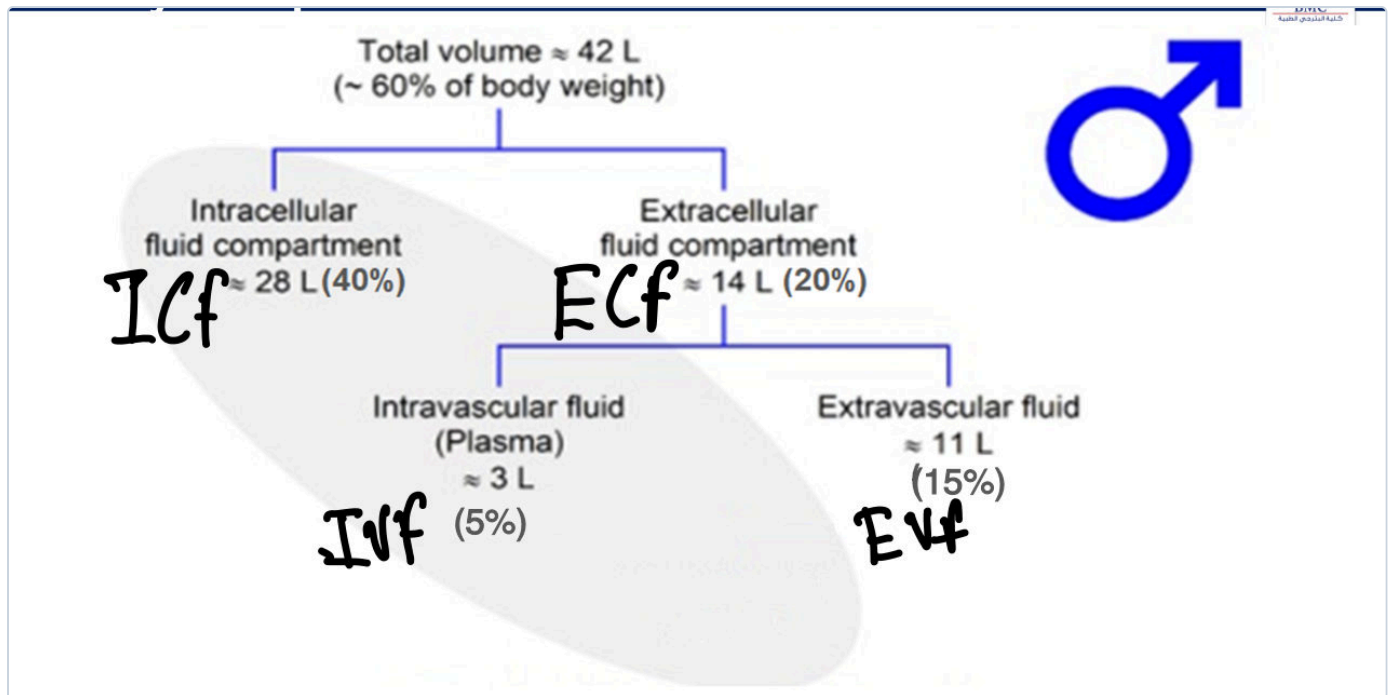


Slide 9: TBW falls from infancy to old age and is lower in women

C. ★★★★★ Compartment volumes, 70-kg adult male (Slide 10)

Compartment	Volume	% body weight	Fraction of TBW [added]
★ Total body water	≈ 42 L	60%	1
★ ICF	≈ 28 L	40%	$\frac{2}{3}$
★ ECF	≈ 14 L	20%	$\frac{1}{3}$
↳ Plasma (intravascular fluid, IVF)	≈ 3 L	5%	$\frac{1}{4}$ of ECF
↳ Interstitial fluid (extravascular, EVF / ISF)	≈ 11 L	15%	$\frac{3}{4}$ of ECF

Mnemonic [added]: "**60-40-20**" (TBW-ICF-ECF), then ECF splits **15 : 5** (ISF : plasma), i.e., **3 : 1**.



Slide 10: Body fluid compartments in the adult male

D. Other facts (Slide 11)

- ★ **Total blood volume ≈ 8% of body weight** (plasma + cells)
- ★ **ICF differs significantly from ECF** in composition
- ★ **Lymph bridges ISF and IVF** (plasma)
- ★ **Total +ve and -ve charges are equal everywhere EXCEPT along the cell membrane surfaces** [added: this local charge separation creates the resting membrane potential]

E. ★★★★★ Ionic composition (Slide 11)

	ICF	Plasma	ISF
Cations (mmol/L)			
★ Na ⁺	14	★ 145	140
★ K ⁺	★ 140	4.5	4.0
Ca ²⁺	0.1 × 10 ⁻³	2.5	1.2
Mg ²⁺	★ 58	1.2	1.2
Anions (mmol/L)			
★ Cl ⁻	3	★ 115	110
HCO ₃ ⁻	10	28	27
★ HPO ₄ ²⁻	★ 75	4	4
Non-charged (mg/dL)			
Glucose	0-20	70-140	70-140
★ Osmolarity (mOsm/L)	★ 290	★ 290	★ 290

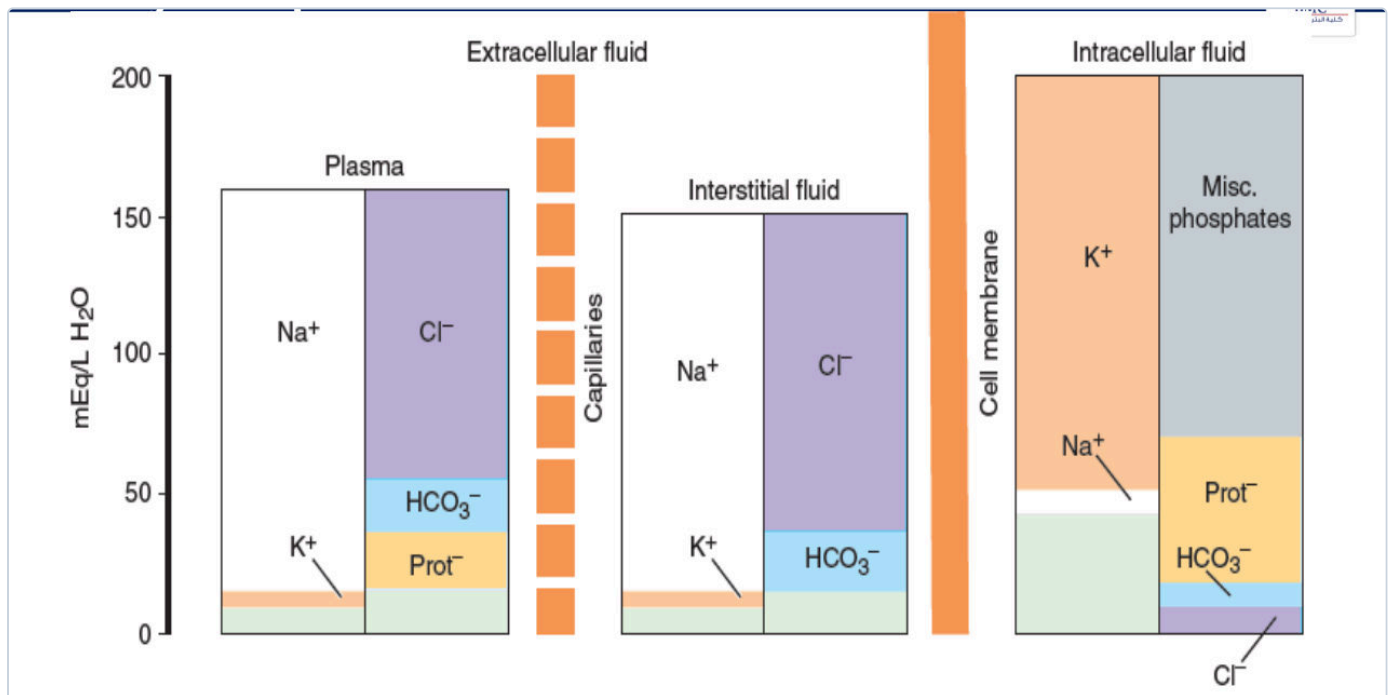
Patterns to learn:

- ★ **ICF**: main cation **K⁺**, main anions **phosphates + proteins**; high **Mg²⁺**
- ★ **ECF**: main cation **Na⁺**, main anions **Cl⁻ + HCO₃⁻**
- ★ **Plasma vs ISF**: almost identical except **protein** (plasma has more, see Slide 12), which is why plasma has slightly more Na⁺ and Cl⁻ [added: Gibbs-Donnan effect]

- ★ **Osmolarity is equal (290) in all compartments:** water moves freely until it is

	ICF	ECF	
		Plasma	ISF
Cations (mmol / L)			
Na ⁺	14	145	140
K ⁺	140	4.5	4.0
Ca ⁺⁺	0.1 x 10 ⁻³	2.5	1.2
Mg ⁺⁺	58	1.2	1.2
Anions (mmol / L)			
Cl ⁻	3	115	110
HCO ₃ ⁻	10	28	27
HPO ₄ ⁻²	75	4	4
Non-charged (mg/dl)			
Glucose	0-20	70-140	70-140
Osmolarity (mOsm/L)	290	290	290

Slide 11: Ionic composition of ICF, plasma and ISF



Slide 12: Ionic bar charts: plasma vs interstitial vs intracellular fluid

F. ★★★★★ Measuring the compartments (Slide 13)

Principle: ★ **Fick's (indicator-dilution) principle** [added formula: $\text{Volume} = \text{amount of marker injected} \div \text{equilibrium concentration (minus any amount excreted/metabolised)}$]

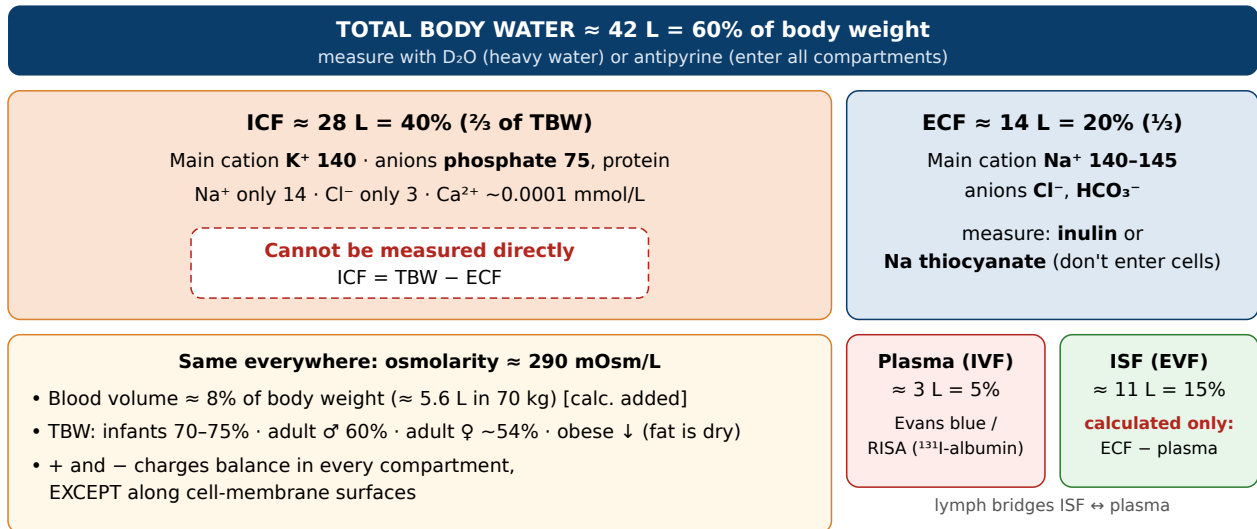
#	Compartment	Method	Marker(s)
1	★ TBW	Measured	★ D₂O (heavy water) or antipyrine (enter all compartments)
2	★ ECF	Measured	★ Inulin or Na thiocyanate (do not enter cells)

#	Compartment	Method	Marker(s)
3	★ Plasma	Measured	★ Evans blue dye or plasma proteins labelled with radioactive iodine
4	★ ICF	Calculated	TBW – ECF
5	★ ISF	Calculated	ECF – plasma volume

📝 **Lecturer's handwritten note on the slide:** ★ **"The ICF and ISF cannot be measured; it can be calculated only."**

Mnemonic [added]: **"D-I-E"**: **D**₂O → whole body; **I**nulin → ECF; **E**vans blue → plasma. ICF and ISF are only **subtracted**.

Body water in a 70-kg adult male: the 60-40-20 rule and how each is measured



Added diagram: body water compartments and their markers

🔄 **Recap: Compartments.** 60% water (infants 70-75, women ~54). TBW 42 L = ICF 28 (40%) + ECF 14 (20%). ECF = ISF 11 (15%) + plasma 3 (5%). Blood is 8% of body weight. ICF is K⁺/phosphate/Mg²⁺; ECF is Na⁺/Cl⁻/HCO₃⁻. Osmolarity is 290 everywhere. Measure TBW (D₂O), ECF (inulin) and plasma (Evans blue); calculate ICF and ISF.

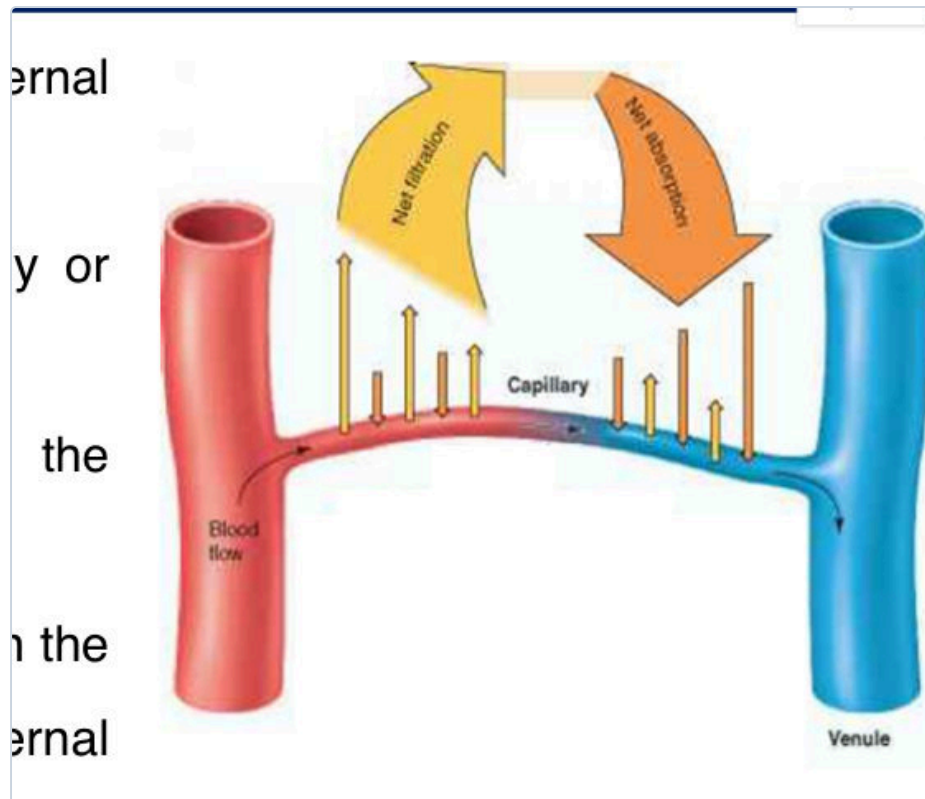
🔗 **Link:** The ISF bathing every cell is the cell's "internal environment". Its composition must stay within narrow limits, and keeping it there is homeostasis.

▶ **Up next: Homeostasis.**

2.3 Homeostasis (Slides 14-18)

★★ **Definition:** keeping the conditions of the **internal environment constant**.

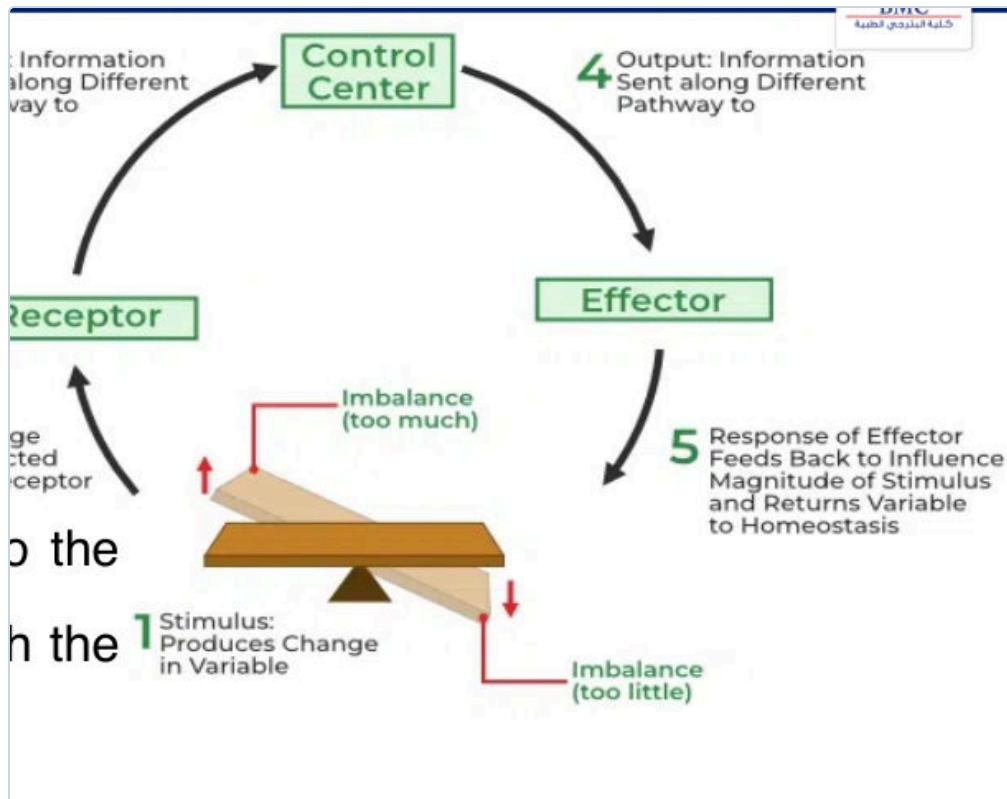
- Most body systems work, **directly or indirectly**, to maintain homeostasis
- ★ Cells are surrounded by **ISF**, which is **the internal environment** for those cells
- Life is compatible only within **narrow limits** of change in the internal environment's chemical/physical properties
- **Examples:** control of **body temperature, pH**, and concentrations of **O₂, CO₂, nutrients, waste products**



Slide 15: Capillary exchange between blood and interstitial fluid (the internal environment)

★★★★ Components of a homeostatic system: "SRACEER" (Slide 16, mnemonic on slide)

#	Letter	Component	Role
1	S	Stimulus	A change in the environment
2	R	Receptor (sensor)	Detects the change
3	A	Afferent (input)	Carries detected information to the centre
4	C	Control centre	Corrects the change back to the ★ set point (the optimum condition for the system)
5	E	Efferent (output)	Carries orders from the centre
6	E	Effectors	Execute the orders
7	R	Response	★ Return to the set point



Slide 16: The homeostatic control loop

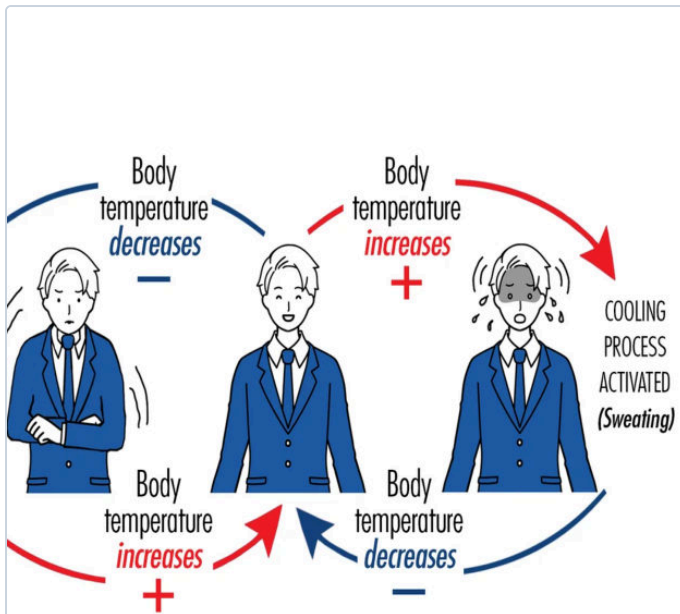
★★★★ Feedback control (Slides 17-18)

Purpose: ★ prevents over- or under-compensation.

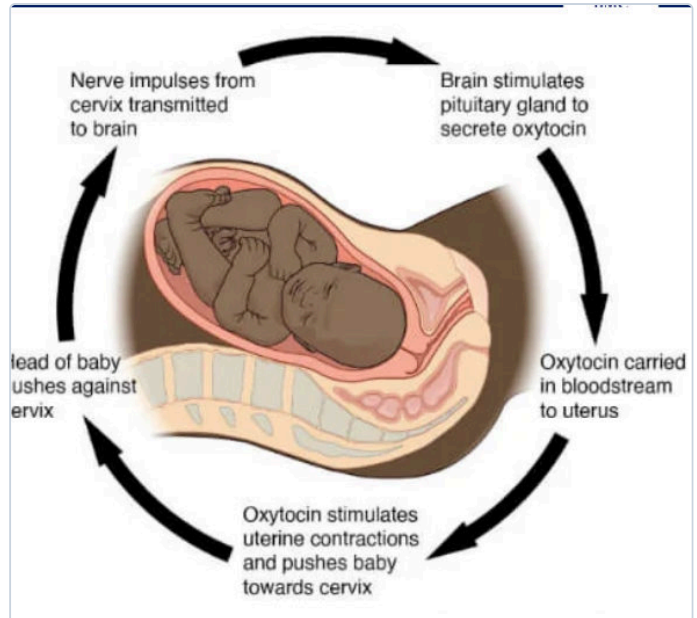
	A. Negative feedback	B. Positive feedback
Frequency	★ Used by most body systems (most common)	Used in some cases (less common)
Effect	★ Reverses the change (response opposes the stimulus)	★ Input increases / accelerates the response (amplifies)
Result	Stability around the set point	A self-amplifying cycle until an end-event [added: e.g., delivery of the baby]
Slide example	★ Body temperature (↑ temp → sweating/cooling → ↓ temp; ↓ temp → heat conservation → ↑ temp)	★ Childbirth / oxytocin
Other examples [added]	BP (baroreceptor reflex), blood glucose (insulin/glucagon), TSH-T4 axis	Blood clotting cascade, LH surge before ovulation, action-potential Na ⁺ influx, milk ejection

★ Oxytocin positive-feedback loop (Slide 18):

1. **Oxytocin** released
2. → ↑ **frequency and strength of uterine contractions**
3. → baby pushed against the **cervix**
4. → **cervical stretch** sends nerve impulses to the brain
5. → **more oxytocin** released from the **pituitary** into the blood
6. → cycle repeats (until delivery [added])



Slide 17: Negative feedback in temperature control



Slide 18: Positive feedback: oxytocin in labour

🔄 **Recap: Homeostasis.** Keep the ISF constant. The loop is SRACEER: stimulus → receptor → afferent → control centre (set point) → efferent → effector → response. Negative feedback (most systems, e.g., temperature) opposes change. Positive feedback (rare, e.g., oxytocin in labour) amplifies it.

🔗 **Link:** The lecture closes with a conclusion slide and one case that tests the compartment markers.

👉 **Up next:** Conclusion + case (\$7).

2.4 Conclusion (Slide 20)

- Physiology = the study of the **normal function of living things**
- Homeostasis = keeping the **composition of the internal environment constant**
- Feedback can be **negative (most common)** or **positive (less common)**
- ★ **Cells have different methods of communication** (*stated in the conclusion but not covered in this lecture's slides, so expect it in a later lecture*)

3. High-Yield Comparison Tables

Table A: Compartment volumes (70-kg male)

	L	% BW	Marker
TBW	42	60	D ₂ O, antipyrine
ICF	28	40	TBW – ECF
ECF	14	20	Inulin, Na thiocyanate
ISF	11	15	ECF – plasma
Plasma	3	5	Evans blue, ¹³¹ I-albumin
Blood	~5.6 [added calc.]	8	

Table B: ICF vs ECF

	ICF	ECF
Main cation	K⁺ (140)	Na⁺ (140-145)
Main anion	Phosphate (75) , protein	Cl⁻ (110-115) , HCO ₃ ⁻
Mg ²⁺	High (58)	Low (1.2)
Ca ²⁺	Tiny (10⁻⁴)	1.2-2.5
Glucose	0-20 mg/dL	70-140 mg/dL
Osmolarity	290	290

Table C: Plasma vs ISF

	Plasma	ISF
Volume	3 L (5%)	11 L (15%)
Protein	High	Low
Na ⁺ / Cl ⁻	145 / 115	140 / 110
Ca ²⁺	2.5	1.2
Separated by	Capillary wall (connected by lymph)	

Table D: Negative vs positive feedback (see §2.3)**Table E: TBW by group (see §2.2 B)**

4. Rules, Exceptions, Patterns

- **60-40-20**: TBW 60%, ICF 40%, ECF 20% of body weight.
- **ECF = $\frac{1}{4}$ plasma + $\frac{3}{4}$ ISF.**
- **TBW falls** from infancy (70-75%) to adult male (60%) to female (~54%) and the obese, because **fat holds little water.**
- **K⁺ inside, Na⁺ outside. Cl⁻ outside, phosphate inside.**
- **Osmolarity is equal across compartments (290)**, even though the ions differ.
- **Electroneutrality holds everywhere EXCEPT at the cell membrane surface.**
- **Only TBW, ECF and plasma are measured. ICF and ISF are calculated.**
- **Marker must stay only in the target space:** inulin can't enter cells (ECF); Evans blue binds albumin and stays in vessels (plasma) *[mechanism added]*.
- **Negative feedback = most systems; positive feedback = exceptions** (oxytocin in labour).
- **Feedback exists to prevent over- or under-compensation.**

5. Clinical Correlations *[added; the lecture has no clinical slide, so these are illustrative]*

- **Infants dehydrate faster** because a larger share of body weight is water (70-75%) and turnover is high.
- **Elderly women and obese patients** have less water reserve and are more prone to dehydration; water-soluble drug dosing differs.

- **Hyperkalaemia or hypokalaemia** disturbs the K^+ gradient (140 inside vs 4 outside) → arrhythmias.
- **Oedema** = expansion of the ISF compartment; plasma protein keeps fluid in vessels.
- **Fever** is a reset of the temperature set point, not a failure of negative feedback.
- **Oxytocin (Syntocinon) infusion** is used to induce/augment labour by harnessing the positive-feedback loop.

6. Rapid Revision (Last-Day)

- Physiology = normal function; human physiology = organs + systems + integration.
- **75-100 trillion cells** → tissues → organs → systems.
- Adult ♂: **60% water, 18% protein, 15% fat, 7% minerals**.
- TBW: **infant 70-75%, ♂ 60%, ♀ ~54%, obese < 60%**.
- **TBW 42 L · ICF 28 L (40%) · ECF 14 L (20%) · ISF 11 L (15%) · plasma 3 L (5%)**.
- **Blood volume ≈ 8% BW**. Lymph links ISF and plasma.
- ICF: **K^+ 140, Na^+ 14, Mg^{2+} 58, HPO_4^{2-} 75, Cl^- 3, HCO_3^- 10, Ca^{2+} 0.1×10^{-3}** .
- Plasma: **Na^+ 145, K^+ 4.5, Ca^{2+} 2.5, Cl^- 115, HCO_3^- 28**. ISF: **Na^+ 140, K^+ 4, Cl^- 110**.
- **Osmolarity 290 everywhere**. Charges balance except at membrane surfaces.
- **Fick principle**. TBW: **D_2O / antipyrine**. ECF: **inulin / Na thiocyanate**. Plasma: **Evans blue / ^{131}I -protein**. **ICF = TBW – ECF. ISF = ECF – plasma** (calculated only).
- Homeostasis = constant internal environment (**ISF**): temperature, pH, O_2 , CO_2 , nutrients, wastes.
- **SRACEER**: Stimulus, Receptor, Afferent, Control centre (set point), Efferent, Effector, Response.
- Feedback prevents over/under-compensation. **Negative = most (temperature); positive = few (oxytocin in labour)**.

7. Exam-Style Questions

Q1 (MCQ, lecture case, Slide 21). A healthy 25-year-old man receives an IV tracer that distributes **only within the ECF**, not entering cells. Its dilution is used to calculate ECF volume. Which tracer? a) Evans blue dye **b) Inulin** c) Deuterium oxide (D_2O) d) ^{131}I -labelled albumin **Answer: b. Inulin** (highlighted on the slide). Inulin (or Na thiocyanate) crosses capillaries but not cell membranes → ECF. **Evans blue and ^{131}I -albumin** stay in the vessels → **plasma**. **D_2O** enters all compartments → **TBW**.

Q2 (MCQ). Which compartment volume **cannot be measured directly** with a single marker? a) Total body water b) Plasma c) ECF **d) Interstitial fluid** **Answer: d.** ISF = **ECF – plasma** (and ICF = TBW – ECF); both are calculated only.

Q3 (MCQ). Which is an example of **positive** feedback per the lecture? a) Sweating when body temperature rises b) Shivering in the cold **c) Oxytocin release during labour** d) Baroreceptor reflex **Answer: c.** Contractions push the baby on the cervix → more oxytocin → stronger contractions (amplification). a, b and d are negative feedback.

Q4 (Calculation). In a 70-kg man, D_2O gives TBW = 42 L and inulin gives ECF = 14 L; Evans blue gives plasma = 3 L. Calculate ICF and ISF. **Answer: ICF = 42 – 14 = 28 L (40% BW). ISF = 14 – 3 = 11 L (15% BW).**

Q5 (Short answer). List the seven components of a homeostatic control system in order. **Answer:** Stimulus → Receptor → Afferent pathway → Control centre (compares with the set point) → Efferent pathway → Effector → Response (return to the set point): **SRACEER**.

8. Exam Pitfalls & Traps

1. **Mixing up markers:** Evans blue = **plasma** (not ECF); inulin = **ECF** (not TBW); D₂O = **TBW**.
2. **Claiming ICF or ISF can be measured directly.** The lecturer's handwritten note says **calculated only**.
3. **ISF vs plasma share of ECF:** ISF is the **bigger** part (15% vs 5%).
4. **Assuming osmolarity differs between ICF and ECF.** The ions differ, but osmolarity is **290 in all**.
5. **"Electroneutrality everywhere."** Correct **except at the membrane surfaces**.
6. **Women vs men TBW:** women are **lower** (~54%) because of more fat, not less muscle water.
7. **Positive feedback is "bad".** It's physiological in labour; it is just **less common**.
8. **⚠ Cell number wording:** the slide says "75–100 trillion (10¹²)". A trillion is 10¹², so the total is ≈ **7.5–10 × 10¹³**. Quote "75–100 trillion".
9. **⚠ Female TBW:** the text says **~54%**, the figure shows **50%**. Use **~54%** (text) unless asked about the figure.
10. **Conclusion mentions "cell communication"** but this lecture doesn't teach it; it's likely the next lecture's topic.